JOINT ADMISSION TEST FOR MASTERS • IIT JAM MATHEMATICS (MA)

IIT JAM 2015 Official Paper

Category: CBT Era (2015–2026) • Official Mock Test Series

TOTAL QUESTIONS	TOTAL MARKS	TEST DURATION	PAPER PATTERN
60	100 M	180 Min	MCQ • MSQ • NAT (60 Ques

Section / Type	Questions	Marking Scheme	Negative Marking	Format / Details
Multiple Choice (MCQ)	30	+1 / +2 Marks	−1/3 / −2/3 Marks	4 Options (1 Correct)
Multiple Select (MSQ)	10	+2 Marks	Nil (0)	4 Options (1+ Correct)
Numerical Answer (NAT)	20	+1 / +2 Marks	Nil (0)	Real Decimal Value

IMPORTANT INSTRUCTIONS FOR CANDIDATES

1. **Authentic Examination PYQs:** This mock paper contains verified official IIT JAM Mathematics questions and high-resolution problem statements.
2. **Section A (MCQ):** Each question has four choices (A), (B), (C), (D), out of which **only one** is correct. Wrong answers carry negative penalty (1/3 of positive marks).
3. **Section B (MSQ):** Each question has four choices (A), (B), (C), (D), out of which **one or more than one** choice(s) may be correct. Full marks are awarded only if all correct choices and zero incorrect choices are chosen. No partial marks and no negative marks.
4. **Section C (NAT):** Numerical answers are real numbers entered via virtual keyboard. Full credit is awarded if the numerical answer falls within the official accepted interval. No negative marking.
5. **Master Answer Key:** Complete official answers and acceptance bounds are provided at the end of this document.
6. **Online Interactive Simulator:** Practice this test with virtual calculator, timers, and instant scoring analytics at **vaibhavgeometer.github.io**.

IIT JAM Mathematics Mock Test Series • Curated by Vaibhav Geometer

Question 1 (JAM 2015 • Q1 • ID: JAM_2015_Q1) MCQ +1 M Neg: −0.33

Q.1 Suppose N is a normal subgroup of a group G. Which one of the following is true?

(A) If G is an infinite group then G/N is an infinite group
(B) If G is a nonabelian group then G/N is a nonabelian group
(C) If G is a cyclic group then G/N is an abelian group
(D) If G is an abelian group then G/N is a cyclic group

Question 2 (JAM 2015 • Q2 • ID: JAM_2015_Q2) MCQ +1 M Neg: −0.33

Q.2 Let $y(x) = u(x)\sin x + v(x)\cos x$ be a solution of the differential equation $y'' + y = \sec x$. Then $u(x)$ is

(A) $\ln|\cos x| + C$
(B) $-x + C$
(C) $x + C$
(D) $\ln|\sec x| + C$

Question 3 (JAM 2015 • Q3 • ID: JAM_2015_Q3) MCQ +1 M Neg: −0.33

Q.3 Let a, b, c, d be distinct non-zero real numbers with $a + b = c + d$. Then an eigenvalue of the matrix $\begin{bmatrix} a & b & 1 \\ c & d & 1 \\ 1 & -1 & 0 \end{bmatrix}$ is

(A) $a + c$
(B) $a + b$
(C) $a - b$
(D) $b - d$

Question 4 (JAM 2015 • Q4 • ID: JAM_2015_Q4) MCQ +1 M Neg: −0.33

Q.4 Let S be a nonempty subset of $\mathbb{R}$. If S is a finite union of disjoint bounded intervals, then which one of the following is true?

(A) If S is not compact, then $\sup S \notin S$ and $\inf S \notin S$
(B) Even if $\sup S \in S$ and $\inf S \in S$, S need not be compact
(C) If $\sup S \in S$ and $\inf S \in S$, then S is compact
(D) Even if S is compact, it is not necessary that $\sup S \in S$ and $\inf S \in S$

Question 5 (JAM 2015 • Q5 • ID: JAM_2015_Q5) MCQ +1 M Neg: −0.33

Q.5 Let $\{x_n\}$ be a convergent sequence of real numbers. If $x_1 > \pi + \sqrt{2}$ and $x_{n+1} = \pi + \sqrt{x_n - \pi}$ for $n \geq 1$, then which one of the following is the limit of this sequence?

(A) $\pi + 1$ (B) $\pi + \sqrt{2}$ (C) π (D) $\pi + \sqrt{\pi}$

Question 6 (JAM 2015 • Q6 • ID: JAM_2015_Q6) MCQ +1 M Neg: −0.33

Q.6 The volume of the portion of the solid cylinder $x^2 + y^2 \leq 2$ bounded above by the surface $z = x^2 + y^2$ and bounded below by the xy-plane is

(A) π (B) 2π (C) 3π (D) 4π

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Question 7 (JAM 2015 • Q7 • ID: JAM_2015_Q7) MCQ +1 M Neg: −0.33

Q.7 Let $f: \mathbb{R} \to \mathbb{R}$ be a differentiable function with $f(0) = 0$. If for all $x \in \mathbb{R}$, $1 < f'(x) < 2$, then which one of the following statements is true on $(0, \infty)$?

(A) f is unbounded
(B) f is increasing and bounded
(C) f has at least one zero
(D) f is periodic

Question 8 (JAM 2015 • Q8 • ID: JAM_2015_Q8) MCQ +1 M Neg: −0.33

Q.8 If an integral curve of the differential equation $(y - x)\frac{dy}{dx} = 1$ passes through $(0, 0)$ and $(\alpha, 1)$, then α is equal to

(A) $2 - e^{-1}$ (B) $1 - e^{-1}$ (C) e^{-1} (D) $1 + e$

Question 9 (JAM 2015 • Q9 • ID: JAM_2015_Q9) MCQ +1 M Neg: −0.33

Q.9 An integrating factor of the differential equation

$$\frac{dy}{dx} = \frac{2xy^2 + y}{x - 2y^3}$$

is

(A) $\frac{1}{y}$ (B) $\frac{1}{y^2}$ (C) y (D) y^2

Question 10 (JAM 2015 • Q10 • ID: JAM_2015_Q10) MCQ +1 M Neg: −0.33

Q.10 Let A be a nonempty subset of $\mathbb{R}$. Let $I(A)$ denote the set of interior points of A. Then $I(A)$ can be

(A) empty
(B) singleton
(C) a finite set containing more than one element
(D) countable but not finite

Q. 11 – Q. 30 carry two marks each.

Question 11 (JAM 2015 • Q11 • ID: JAM_2015_Q11) MCQ +2 M Neg: −0.67

Q.11 Let S_3 be the group of permutations of three distinct symbols. The direct sum $S_3 \oplus S_3$ has an element of order

(A) 4 (B) 6 (C) 9 (D) 18

Question 12 (JAM 2015 • Q12 • ID: JAM_2015_Q12) MCQ +2 M Neg: −0.67

Q.12 The orthogonal trajectories of the family of curves $y = C_1 x^3$ are

(A) $2\,x^2 + 3\,y^2 = C_2$ (B) $3\,x^2 + y^2 = C_2$
(C) $3\,x^2 + 2\,y^2 = C_2$ (D) $x^2 + 3\,y^2 = C_2$

Question 13 (JAM 2015 • Q13 • ID: JAM_2015_Q13) MCQ +2 M Neg: −0.67

Q.13 Let G be a nonabelian group. Let $\alpha \in G$ have order 4 and let $\beta \in G$ have order 3. Then the order of the element $\alpha\beta$ in G

(A) is 6
(B) is 12
(C) is of the form $12k$ for $k \geq 2$
(D) need not be finite

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Question 14 (JAM 2015 • Q14 • ID: JAM_2015_Q14) MCQ +2 M Neg: −0.67

Q.14 Let S be the bounded surface of the cylinder $x^2 + y^2 = 1$ cut by the planes $z = 0$ and $z = 1 + x$. Then the value of the surface integral $\iint_S 3z^2 d\sigma$ is equal to

(A) $\int_0^{2\pi} (1 + \cos\theta)^3 d\theta$
(B) $\int_0^{2\pi} \sin\theta \cos\theta \,(1 + \cos\theta)^2 d\theta$
(C) $\int_0^{2\pi} (1 + 2\cos\theta)^3 d\theta$
(D) $\int_0^{2\pi} \sin\theta \cos\theta \,(1 + 2\cos\theta)^2 d\theta$

Question 15 (JAM 2015 • Q15 • ID: JAM_2015_Q15) MCQ +2 M Neg: −0.67

Q.15 Suppose that the dependent variables z and w are functions of the independent variables x and y, defined by the equations $f(x, y, z, w) = 0$ and $g(x, y, z, w) = 0$, where $f_z g_w - f_w g_z = 1$. Which one of the following is correct?

(A) $z_x = f_w g_x - f_x g_w$
(B) $z_x = f_x g_w - f_w g_x$
(C) $z_x = f_z g_x - f_x g_z$
(D) $z_x = f_z g_w - f_z g_x$

Question 16 (JAM 2015 • Q16 • ID: JAM_2015_Q16) MCQ +2 M Neg: −0.67

Q.16

Let $A = \begin{bmatrix} 0 & 1-i \\ -1-i & i \end{bmatrix}$ and $B = A^T \overline{A}$. Then

(A) an eigenvalue of B is purely imaginary
(B) an eigenvalue of A is zero
(C) all eigenvalues of B are real
(D) A has a non-zero real eigenvalue

Question 17 (JAM 2015 • Q17 • ID: JAM_2015_Q17) MCQ +2 M Neg: −0.67

Q.17 The limit

$$\lim_{x \to 0+} \frac{1}{\sin^2 x} \int_{\frac{x}{2}}^{x} \sin^{-1} t \, dt$$

is equal to

(A) 0 (B) $\frac{1}{8}$ (C) $\frac{1}{4}$ (D) $\frac{3}{8}$

Question 18 (JAM 2015 • Q18 • ID: JAM_2015_Q18) MCQ +2 M Neg: −0.67

Q.18 Let $P_2(\mathbb{R})$ be the vector space of polynomials in x of degree at most 2 with real coefficients. Let $M_2(\mathbb{R})$ be the vector space of 2×2 real matrices. If a linear transformation $T: P_2(\mathbb{R}) \to M_2(\mathbb{R})$ is defined as

$$T(f) = \begin{bmatrix} f(0) - f(2) & 0 \\ 0 & f(1) \end{bmatrix}$$

then

(A) T is one-one but not onto
(B) T is onto but not one-one
(C) Range (T) = span $\left\{ \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}, \begin{bmatrix} -2 & 0 \\ 0 & 1 \end{bmatrix} \right\}$
(D) Null (T) = span $\{x^2 - 2x, \ 1 - x\}$

Question 19 (JAM 2015 • Q19 • ID: JAM_2015_Q19) MCQ +2 M Neg: −0.67

Q.19

Let $B_1 = \{(1,2),\ (2,-1)\}$ and $B_2 = \{(1,0),\ (0,1)\}$ be ordered bases of $\mathbb{R}^2$. If $T:\ \mathbb{R}^2 \to \mathbb{R}^2$ is a linear transformation such that $[T]_{B_1,B_2}$, the matrix of T with respect to B_1 and B_2, is $\begin{bmatrix} 4 & 3 \\ 2 & -4 \end{bmatrix}$, then $T(5,5)$ is equal to

(A) $(-9,\ 8)$ (B) $(9,\ 8)$ (C) $(-15,\ -2)$ (D) $(15,\ 2)$

Question 20 (JAM 2015 • Q20 • ID: JAM_2015_Q20) MCQ +2 M Neg: −0.67

Q.20

Let $S = \bigcap_{n=1}^{\infty}\left(\left[0,\ \frac{1}{2n+1}\right] \cup \left[\frac{1}{2n},\ 1\right]\right)$. Which one of the following statements is **FALSE**?

(A) There exist sequences $\{a_n\}$ and $\{b_n\}$ in [0,1] such that $S = [0,1] \setminus \bigcup_{n=1}^{\infty}(a_n,\ b_n)$
(B) $[0,1]\setminus S$ is an open set
(C) If A is an infinite subset of S, then A has a limit point
(D) There exists an infinite subset of S having no limit points

Question 21 (JAM 2015 • Q21 • ID: JAM_2015_Q21) MCQ +2 M Neg: −0.67

Q.21 Let $f\colon \mathbb{R} \to \mathbb{R}$ be a strictly increasing continuous function. If $\{a_n\}$ is a sequence in $[0,1]$, then the sequence $\{f(a_n)\}$ is

(A) increasing (B) bounded
(C) convergent (D) not necessarily bounded

Question 22 (JAM 2015 • Q22 • ID: JAM_2015_Q22) MCQ +2 M Neg: −0.67

Q.22

Which one of the following statements is true for the series $\sum_{n=1}^{\infty}(-1)^n \frac{(2n)!}{n^{2n}}$?

(A) The series converges conditionally but not absolutely
(B) The series converges absolutely
(C) The sequence of partial sums of the series is bounded but not convergent
(D) The sequence of partial sums of the series is unbounded

Question 23 (JAM 2015 • Q23 • ID: JAM_2015_Q23) MCQ +2 M Neg: −0.67

Q.23

The sequence $\left\{\cos\left(\frac{1}{2}\tan^{-1}\left(-\frac{n}{2}\right)^n\right)\right\}$ is

(A) monotone and convergent
(B) monotone but not convergent
(C) convergent but not monotone
(D) neither monotone nor convergent

Question 24 (JAM 2015 • Q24 • ID: JAM_2015_Q24) MCQ +2 M Neg: −0.67

Q.24

If $y(t)$ is a solution of the differential equation $y'' + 4y = 2e^t$, then

$$\lim_{t\to\infty} e^{-t}\, y(t)$$

is equal to

(A) $\frac{2}{3}$ (B) $\frac{2}{5}$ (C) $\frac{2}{7}$ (D) $\frac{2}{9}$

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Question 25 (JAM 2015 • Q25 • ID: JAM_2015_Q25) MCQ +2 M Neg: −0.67

Q.25

For what real values of x and y, does the integral $\int_x^y (6 - t - t^2)dt$ attain its maximum?

(A) $x = -3,\ y = 2$ (B) $x = 2,\ y = 3$
(C) $x = -2,\ y = 2$ (D) $x = -3,\ y = 4$

Question 26 (JAM 2015 • Q26 • ID: JAM_2015_Q26) MCQ +2 M Neg: −0.67

Q.26 The area of the planar region bounded by the curves $x = 6y^2 - 2$ and $x = 2y^2$ is

(A) $\frac{\sqrt{2}}{3}$ (B) $\frac{2\sqrt{2}}{3}$ (C) $\frac{4\sqrt{2}}{3}$ (D) $\sqrt{2}$

Question 27 (JAM 2015 • Q27 • ID: JAM_2015_Q27) MCQ +2 M Neg: −0.67

Q.27

For $n \geq 2$, let $f_n: \mathbb{R} \to \mathbb{R}$ be given by $f_n(x) = x^n \sin x$. Then at $x = 0$, f_n has a

(A) local maximum if n is even

(B) local maximum if n is odd

(C) local minimum if n is even

(D) local minimum if n is odd

Question 28 (JAM 2015 • Q28 • ID: JAM_2015_Q28) MCQ +2 M Neg: −0.67

Q.28

For $m, n \in \mathbb{N}$, define $f_{m,n}(x) = \begin{cases} x^m \sin\left(\frac{1}{x^n}\right), & x \neq 0 \\ 0 & x = 0 \end{cases}$

Then at $x = 0$, $f_{m,n}$ is

(A) differentiable for each pair m, n with $m > n$

(B) differentiable for each pair m, n with $m < n$

(C) not differentiable for each pair m, n with $m > n$

(D) not differentiable for each pair m, n with $m < n$

Question 29 (JAM 2015 • Q29 • ID: JAM_2015_Q29) MCQ +2 M Neg: −0.67

Q.29

Let G and H be nonempty subsets of $\mathbb{R}$, where G is connected and $G \cup H$ is not connected. Which one of the following statements is true for all such G and H ?

(A) If $G \cap H = \emptyset$, then H is connected

(B) If $G \cap H = \emptyset$, then H is not connected

(C) If $G \cap H \neq \emptyset$, then H is connected

(D) If $G \cap H \neq \emptyset$, then H is not connected

Question 30 (JAM 2015 • Q30 • ID: JAM_2015_Q30) MCQ +2 M Neg: −0.67

Q.30

Let $f : \{ (x,y) \in \mathbb{R}^2 : x > 0,\ y > 0 \} \to \mathbb{R}$ be given by

$$f(x,y) = x^{\frac{1}{3}} y^{\frac{-4}{3}} \tan^{-1}\left(\frac{y}{x}\right) + \frac{1}{\sqrt{x^2 + y^2}}$$

Then the value of

$$g(x,y) = \frac{x f_x(x,y) + y f_y(x,y)}{f(x,y)}$$

(A) changes with x but not with y

(B) changes with y but not with x

(C) changes with x and also with y

(D) neither changes with x nor with y

SECTION - B

MULTIPLE SELECT QUESTIONS (MSQ)

Q. 1 – Q. 10 carry two marks each.

Question 31 (JAM 2015 • Q31 • ID: JAM_2015_Q31) MSQ +2 M Neg: Nil (0)

Q.1

Let $f: \mathbb{R} \to \mathbb{R}$ be a function defined by $f(x) = \int_{-5}^{x} (t-1)^3 dt$.
In which of the following interval(s), f takes the value 1?

(A) $[-6,\ 0]$ (B) $[-2,\ 4]$ (C) $[2,\ 8]$ (D) $[6,\ 12]$

Question 32 (JAM 2015 • Q32 • ID: JAM_2015_Q32) MSQ +2 M Neg: Nil (0)

Q.2

Which of the following statements is (are) true?

(A) $\mathbb{Z}_2 \oplus \mathbb{Z}_3$ is isomorphic to $\mathbb{Z}_6$

(B) $\mathbb{Z}_3 \oplus \mathbb{Z}_3$ is isomorphic to $\mathbb{Z}_9$

(C) $\mathbb{Z}_4 \oplus \mathbb{Z}_6$ is isomorphic to $\mathbb{Z}_{24}$

(D) $\mathbb{Z}_2 \oplus \mathbb{Z}_3 \oplus \mathbb{Z}_5$ is isomorphic to $\mathbb{Z}_{30}$

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Question 33 (JAM 2015 • Q33 • ID: JAM_2015_Q33) MSQ +2 M Neg: Nil (0)

Q.3

Which of the following conditions implies (imply) the convergence of a sequence $\{x_n\}$ of real numbers?

(A) Given $\varepsilon > 0$ there exists an $n_0 \in \mathbb{N}$ such that for all $n \geq n_0$, $|x_{n+1} - x_n| < \varepsilon$

(B) Given $\varepsilon > 0$ there exists an $n_0 \in \mathbb{N}$ such that for all $n \geq n_0$, $\frac{1}{(n+1)^2}|x_{n+1} - x_n| < \varepsilon$

(C) Given $\varepsilon > 0$ there exists an $n_0 \in \mathbb{N}$ such that for all $n \geq n_0$, $(n+1)^2|x_{n+1} - x_n| < \varepsilon$

(D) Given $\varepsilon > 0$ there exists an $n_0 \in \mathbb{N}$ such that for all m, n with $m > n \geq n_0$, $|x_m - x_n| < \varepsilon$

Question 34 (JAM 2015 • Q34 • ID: JAM_2015_Q34) MSQ +2 M Neg: Nil (0)

Q.4

Let $\vec{F}$ be a vector field given by $\vec{F}(x,y,z) = -y\,\hat{\imath} + 2\,x\,y\,\hat{\jmath} + z^3\hat{k}$, for $(x,y,z) \in \mathbb{R}^3$. If C is the curve of intersection of the surfaces $x^2 + y^2 = 1$ and $y + z = 2$, then which of the following is (are) equal to $\left|\int_C \vec{F}\,.\,d\vec{r}\right|$?

(A) $\int_0^{2\pi}\int_0^1 (1 + 2r\sin\theta) r\,dr d\theta$

(B) $\int_0^{2\pi}\left(\frac{1}{2} + \frac{2}{3}\sin\theta\right)\,d\theta$

(C) $\int_0^{2\pi}\int_0^1 (1 + 2r\sin\theta)\,\,dr d\theta$

(D) $\int_0^{2\pi}(1 + \sin\theta)\,d\theta$

Question 35 (JAM 2015 • Q35 • ID: JAM_2015_Q35) MSQ +2 M Neg: Nil (0)

Q.5

Let V be the set of 2×2 matrices $\begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}$ with complex entries such that $a_{11} + a_{22} = 0$. Let W be the set of matrices in V with $a_{12} + \overline{a_{21}} = 0$. Then, under usual matrix addition and scalar multiplication, which of the following is (are) true?

(A) V is a vector space over $\mathbb{C}$
(B) W is a vector space over $\mathbb{C}$
(C) V is a vector space over $\mathbb{R}$
(D) W is a vector space over $\mathbb{R}$

Question 36 (JAM 2015 • Q36 • ID: JAM_2015_Q36) MSQ +2 M Neg: Nil (0)

Q.6

The initial value problem

$$y' = \sqrt{y},\; y(0) = \alpha,\;\; \alpha \geq 0$$

has

(A) at least two solutions if $\alpha = 0$

(B) no solution if $\alpha > 0$

(C) at least one solution if $\alpha > 0$

(D) a unique solution if $\alpha = 0$

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Question 37 (JAM 2015 • Q37 • ID: JAM_2015_Q37) MSQ +2 M Neg: Nil (0)

Q.7 Which of the following statements is (are) true on the interval $\left(0, \frac{\pi}{2}\right)$?

(A) $\cos x < \cos(\sin x)$

(B) $\tan x < x$

(C) $\sqrt{1+x} < 1 + \frac{x}{2} - \frac{x^2}{8}$

(D) $\frac{1-x^2}{2} < \ln(2+x)$

Question 38 (JAM 2015 • Q38 • ID: JAM_2015_Q38) MSQ +2 M Neg: Nil (0)

Q.8 Let $f: \mathbb{R}^2 \longrightarrow \mathbb{R}$ be defined by

$$f(x,y) = \begin{cases} xy\dfrac{x^2-y^2}{x^2+y^2}, & (x,y) \neq (0,0) \\ 0, & (x,y) = (0,0) \end{cases}$$

At $(0,0)$,

(A) f is not continuous

(B) f is continuous, and both f_x and f_y exist

(C) f is differentiable

(D) f_x and f_y exist but f is not differentiable

Question 39 (JAM 2015 • Q39 • ID: JAM_2015_Q39) MSQ +2 M Neg: Nil (0)

Q.9 Let $f, g : [0,1] \to [0,1]$ be functions. Let $R(f)$ and $R(g)$ be the ranges of f and g, respectively. Which of the following statements is (are) true?

(A) If $f(x) \le g(x)$ for all $x \in [0,1]$, then $\sup R(f) \le \inf R(g)$

(B) If $f(x) \le g(x)$ for some $x \in [0,1]$, then $\inf R(f) \le \sup R(g)$

(C) If $f(x) \le g(y)$ for some $x, y \in [0,1]$, then $\inf R(f) \le \sup R(g)$

(D) If $f(x) \le g(y)$ for all $x, y \in [0,1]$, then $\sup R(f) \le \inf R(g)$

Question 40 (JAM 2015 • Q40 • ID: JAM_2015_Q40) MSQ +2 M Neg: Nil (0)

Q.10

Let $f: (-1, 1) \longrightarrow \mathbb{R}$ be the function defined by

$$f(x) = x^2 e^{1/(1-x^2)}$$

Then

(A) f is decreasing in $(-1, 0)$

(B) f is increasing in $(0, 1)$

(C) $f(x) = 1$ has two solutions in $(-1, 1)$

(D) $f(x) = 1$ has no solutions in $(-1, 1)$

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Question 41 (JAM 2015 • Q41 • ID: JAM_2015_Q41) NAT +1 M Neg: Nil (0)

Q. 1 – Q. 10 carry one mark each.

Q.1

Let C be the straight line segment from $P(0, \pi)$ to $Q\left(4, \frac{\pi}{2}\right)$, in the xy-plane. Then the value of $\int_C e^x(\cos y\, dx - \sin y\, dy)$ is ______________

Question 42 (JAM 2015 • Q42 • ID: JAM_2015_Q42) NAT +1 M Neg: Nil (0)

Q.2

Let S be the portion of the surface $z = \sqrt{16 - x^2}$ bounded by the planes $x = 0$, $x = 2$, $y = 0$, and $y = 3$. The surface area of S, correct upto three decimal places, is ______________

Question 43 (JAM 2015 • Q43 • ID: JAM_2015_Q43) NAT +1 M Neg: Nil (0)

Q.3

The number of distinct normal subgroups of S_3 is ______________

Question 44 (JAM 2015 • Q44 • ID: JAM_2015_Q44) NAT +1 M Neg: Nil (0)

Q.4

Let $f\colon \mathbb{R}^2 \longrightarrow \mathbb{R}$ be defined by

$$f(x,y) = \begin{cases} \left(1+\dfrac{x}{y}\right)^2, & y \neq 0 \\ 0, & y = 0 \end{cases}$$

If the directional derivative of f at $(0,0)$ exists along the direction $\cos\alpha\ \hat{\imath} + \sin\alpha\ \hat{\jmath}$, where $\sin\alpha \neq 0$, then the value of $\cot\alpha$ is ______________

Question 45 (JAM 2015 • Q45 • ID: JAM_2015_Q45) NAT +1 M Neg: Nil (0)

Q.5

Let $f\colon \mathbb{R}^3 \longrightarrow \mathbb{R}$ be defined by

$$f(x,y,z) = \sin x + 2\,e^{\frac{y}{2}} + z^2$$

The maximum rate of change of f at $\left(\frac{\pi}{4},\ 0,\ 1\right)$, correct upto three decimal places, is ______________

Question 46 (JAM 2015 • Q46 • ID: JAM_2015_Q46) NAT +1 M Neg: Nil (0)

Q.6

If the power series

$$\sum_{n=0}^{\infty} \frac{n!}{n^n} x^{2n}$$

converges for $|x| < c$ and diverges for $|x| > c$, then the value of c, correct upto three decimal places, is ______________

Question 47 (JAM 2015 • Q47 • ID: JAM_2015_Q47) NAT +1 M Neg: Nil (0)

Q.7

If $5^{2015} \equiv n \pmod{11}$ and $n \in \{0,1,2,3,4,5,6,7,8,9,10\}$, then n is equal to _________

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Question 48 (JAM 2015 • Q48 • ID: JAM_2015_Q48) NAT +1 M Neg: Nil (0)

Q.8

If the set $\left\{\begin{bmatrix} x & -x \\ -1 & 0 \end{bmatrix}, \begin{bmatrix} 0 & -1 \\ x & x \end{bmatrix}, \begin{bmatrix} 1 & -1 \\ 1 & 0 \end{bmatrix}\right\}$ is linearly dependent in the vector space of all 2×2 matrices with real entries, then x is equal to ______________

Question 49 (JAM 2015 • Q49 • ID: JAM_2015_Q49) NAT +1 M Neg: Nil (0)

Q.9

Let $f: \mathbb{R} \longrightarrow \mathbb{R}$ be defined by

$$f(x) = \begin{cases} x^6 - 1, & x \in \mathbb{Q} \\ 1 - x^6, & x \notin \mathbb{Q} \end{cases}$$

The number of points at which f is continuous, is ______________

Question 50 (JAM 2015 • Q50 • ID: JAM_2015_Q50) NAT +1 M Neg: Nil (0)

Q.10

Let $f: (0,1) \to \mathbb{R}$ be a continuously differentiable function such that f' has finitely many zeros in $(0,1)$ and f' changes sign at exactly two of these points. Then for any $y \in \mathbb{R}$, the maximum number of solutions to $f(x) = y$ in $(0,1)$ is ______________

Q. 11 – Q. 20 carry two marks each

Question 51 (JAM 2015 • Q51 • ID: JAM_2015_Q51) NAT +2 M Neg: Nil (0)

Q. 11 – Q. 20 carry two marks each.

Q.11 Let R be the planar region bounded by the lines $x = 0$, $y = 0$ and the curve $x^2 + y^2 = 4$, in the first quadrant. Let C be the boundary of R, oriented counter-clockwise. Then the value of

$$\oint_C x(1-y)\,dx + (x^2 - y^2)\,dy$$

is ______________

Question 52 (JAM 2015 • Q52 • ID: JAM_2015_Q52) NAT +2 M Neg: Nil (0)

Q.12 Suppose G is a cyclic group and $\sigma, \tau \in G$ are such that order(σ) = 12 and order(τ) = 21. Then the order of the smallest group containing σ and τ is ______________

Question 53 (JAM 2015 • Q53 • ID: JAM_2015_Q53) NAT +2 M Neg: Nil (0)

Q.13 The limit

$$\lim_{n\to\infty} \sum_{k=2}^{n} \frac{1}{k^3 - k}$$

is equal to ______________

Question 54 (JAM 2015 • Q54 • ID: JAM_2015_Q54) NAT +2 M Neg: Nil (0)

Q.14 Let $M_2(\mathbb{R})$ be the vector space of 2×2 real matrices. Let V be a subspace of $M_2(\mathbb{R})$ defined by

$$V = \left\{ A \in M_2(\mathbb{R}) : A \begin{bmatrix} 0 & 2 \\ 3 & 1 \end{bmatrix} = \begin{bmatrix} 0 & 2 \\ 3 & 1 \end{bmatrix} A \right\}$$

Then the dimension of V is ______________

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Question 55 (JAM 2015 • Q55 • ID: JAM_2015_Q55) NAT +2 M Neg: Nil (0)

Q.15

Let $f: \mathbb{R}^2 \to \mathbb{R}$ be defined by

$$f(x,y) = \begin{cases} \dfrac{y}{\sin y}, & y \neq 0 \\ 1, & y = 0 \end{cases}$$

Then the integral

$$\frac{1}{\pi^2} \int_{x=0}^{1} \int_{y=\sin^{-1} x}^{\frac{\pi}{2}} f(x,y)\, dy\, dx$$

correct upto three decimal places, is ______________

Question 56 (JAM 2015 • Q56 • ID: JAM_2015_Q56) NAT +2 M Neg: Nil (0)

Q.16

The coefficient of $\left(x-\frac{\pi}{4}\right)^3$ in the Taylor series expansion of the function

$$f(x) = 3\sin x\cos\left(x+\frac{\pi}{4}\right), \qquad x \in \mathbb{R}$$

about the point $\frac{\pi}{4}$, correct upto three decimal places, is ______________

Question 57 (JAM 2015 • Q57 • ID: JAM_2015_Q57) NAT +2 M Neg: Nil (0)

Q.17

If $\int_0^x \left(e^{-t^2}+\cos t\right)\, dt$ has the power series expansion $\sum_{n=1}^{\infty} a_n x^n$, then a_5, correct upto three decimal places, is equal to ______________

Question 58 (JAM 2015 • Q58 • ID: JAM_2015_Q58) NAT +2 M Neg: Nil (0)

Q.18

Let ℓ be the length of the portion of the curve $x = x(y)$ between the lines $y = 1$ and $y = 3$, where $x(y)$ satisfies

$$\frac{dx}{dy} = \frac{\sqrt{1+y^2+y^4}}{y}, \qquad x(1) = 0$$

The value of ℓ, correct upto three decimal places, is ______________

Question 59 (JAM 2015 • Q59 • ID: JAM_2015_Q59) NAT +2 M Neg: Nil (0)

Q.19 The limit

$$\lim_{x\to 0+} \frac{9}{x}\left(\frac{1}{\tan^{-1} x} - \frac{1}{x}\right)$$

is equal to ______________

Question 60 (JAM 2015 • Q60 • ID: JAM_2015_Q60) NAT +2 M Neg: Nil (0)

Q.20 Let P and Q be two real matrices of size 4×6 and 5×4, respectively. If $\text{rank}(Q) = 4$ and $\text{rank}(QP) = 2$, then $\text{rank}(P)$ is equal to ______________

END OF THE QUESTION PAPER

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OFFICIAL MASTER ANSWER KEY & EVALUATION RUBRIC

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Q. No.	Type	Marks	Official Key
1	MCQ	+1	C
2	MCQ	+1	C
3	MCQ	+1	B
4	MCQ	+1	B
5	MCQ	+1	A
6	MCQ	+1	B
7	MCQ	+1	A
8	MCQ	+1	C
9	MCQ	+1	B
10	MCQ	+1	A
11	MCQ	+2	B
12	MCQ	+2	D
13	MCQ	+2	D
14	MCQ	+2	A
15	MCQ	+2	A
16	MCQ	+2	C
17	MCQ	+2	D
18	MCQ	+2	C
19	MCQ	+2	D
20	MCQ	+2	D

Q. No.	Type	Marks	Official Key
21	MCQ	+2	B
22	MCQ	+2	B
23	MCQ	+2	A
24	MCQ	+2	B
25	MCQ	+2	A
26	MCQ	+2	C
27	MCQ	+2	D
28	MCQ	+2	A
29	MCQ	+2	D
30	MCQ	+2	D
31	MSQ	+2	A;C;D
32	MSQ	+2	A;D
33	MSQ	+2	C;D
34	MSQ	+2	A;B
35	MSQ	+2	A;C;D
36	MSQ	+2	A;C
37	MSQ	+2	A;D
38	MSQ	+2	B;C
39	MSQ	+2	B;C;D
40	MSQ	+2	A;B;C

Q. No.	Type	Marks	Official Key
41	NAT	+1	1
42	NAT	+1	6.28 to 6.29
43	NAT	+1	3
44	NAT	+1	-1
45	NAT	+1	2.34 to 2.35
46	NAT	+1	1.64 to 1.65
47	NAT	+1	1
48	NAT	+1	-1
49	NAT	+1	2
50	NAT	+1	3
51	NAT	+2	8
52	NAT	+2	84
53	NAT	+2	0.25
54	NAT	+2	2
55	NAT	+2	0.125
56	NAT	+2	1.41 to 1.42
57	NAT	+2	0.10 to 0.11
58	NAT	+2	5.09 to 5.10
59	NAT	+2	3
60	NAT	+2	2

SCORING METHODOLOGY & EVALUATION GUIDELINES

- **Multiple Choice Questions (MCQ):** Full marks for correct single choice. Negative marks deducted for each incorrect attempt (-1/3 for 1-mark questions, $-2/3$ for 2-mark questions). Unattempted questions award zero.
- **Multiple Select Questions (MSQ):** Full marks if and only if all correct choices are marked and no incorrect choice is selected. Zero marks for partial selection or any incorrect choice. No negative marking.
- **Numerical Answer Type (NAT):** Any numerical value falling strictly within the specified official range (e.g., [a, b]) receives full credit. No negative marking.
- **Marks to All (MTA):** Questions officially designated as MTA award full marks to all candidates regardless of attempt.